Types of Problems in Problem Solving

Problem Solving is a fundamental skill used in academics, computer science, artificial intelligence, management, and everyday life

Definition of a Problem

A **problem** is a situation in which:

A **desired goal** exists

The **path to reach the goal is not immediately known**

Certain **constraints or limitations** are present

Problem solving is the **systematic process of understanding the problem, identifying possible solutions, and selecting the most effective one**.

Need for Classifying Problems

Why Classify Problems?

Different problems require **different solving approaches**

Helps in selecting:

- Algorithms
- Logical reasoning
- Creativity and decision-making

Widely used in:

- Computer Science
- Artificial Intelligence
- Business and Management

Well-Defined Problems

- A **well-defined problem** is a problem in which **the initial state, goal state, constraints, and solution steps are clearly specified**.
- **Characteristics**
- Problem statement is clear and complete
- All required data is available
- Rules and constraints are fixed
- One correct or optimal solution exists
- **Examples**
- Solving a mathematical equation
- Writing a program to calculate factorial
- Finding the area of a circle

ill-Defined Problems

- An **ill-defined problem** is a problem in which **either the goal, the constraints, or the method of solution is unclear or incomplete**.
- **Characteristics**
- No single correct solution
- Information may be missing
- Requires creativity and judgment
- Solutions vary from person to person
- **Examples**
- How to reduce unemployment?
- How to improve education quality?
- How to increase customer satisfaction?

Complex Problems

- A **complex problem** is a problem that **involves multiple interrelated factors and requires a systematic and multi-step approach**.
- **Characteristics**
- Many variables
- Time-consuming
- Requires planning and analysis
- **Examples**
- Climate change control
- Large software system design
- National economic planning

Decision-Making Problems

A **decision-making problem** is a problem in which **one option must be selected from several alternatives based on constraints and expected outcomes**.

Characteristics

Involves choice and risk

Requires evaluation of alternatives

Often time-sensitive

Examples

Choosing a career

Selecting an investment plan

Buying a house

Optimization Problems

- An **optimization problem** is a problem in which the objective is to **maximize or minimize a specific performance measure under given constraints**.
- **Characteristics**
- Clear objective function
- Presence of constraints
- Widely used in AI and operations research
- **Examples**
- Minimizing production cost
- Maximizing profit
- Shortest path in navigation systems

Simple Problems

- A **simple problem** is a problem that **involves few variables and can be solved using basic logic or calculations**.
- **Characteristics**
- Easy to understand
- Short solution time
- Low complexity
- **Examples**
- Adding two numbers
- Finding average marks
- Unit conversions

Structured Problems

A **structured problem** is a problem that can be solved using **predefined rules, procedures, or algorithms**.

Characteristics

Repetitive and predictable

Easily computerized

Clear inputs and outputs

Examples

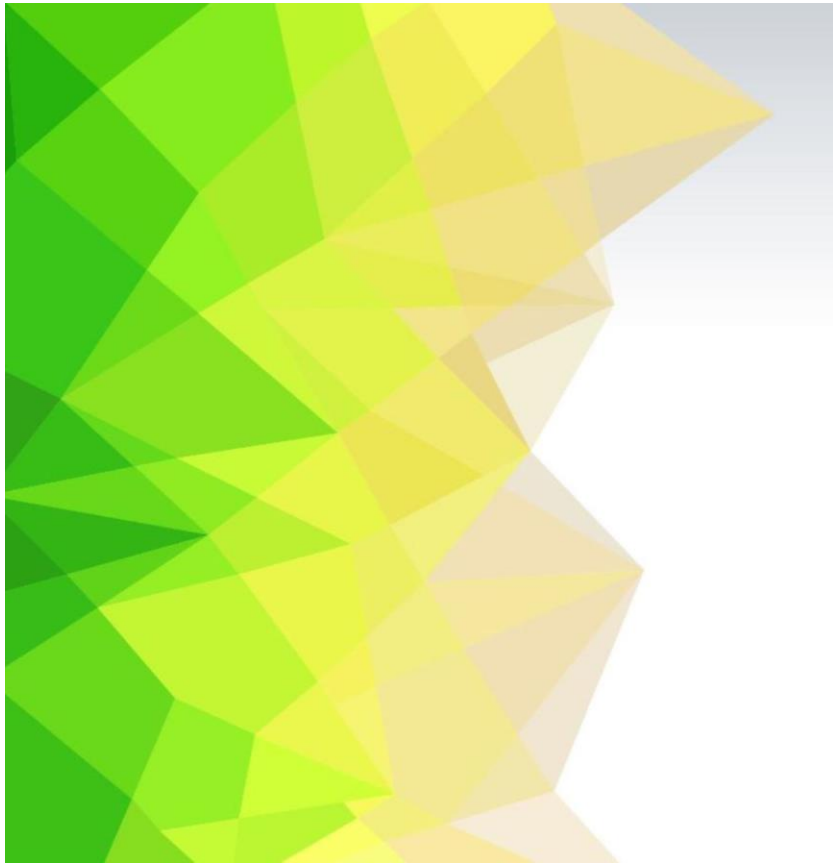
Salary calculation

Attendance processing

Bank interest computation

Unstructured Problems

- An **unstructured problem** is a problem that **does not have a standard solution procedure and requires human insight and experience**.
- **Characteristics**
- Complex and unique
- High uncertainty
- Strategic in nature
- **Examples**
- Business strategy formulation
- Crisis management
- Designing a new product



Routine Problems

- **Definition**
- A **routine problem** is a problem that **occurs frequently and can be solved using previously known methods.**
- **Characteristics**
- Familiar in nature
- Low risk
- Minimal decision making required
- **Examples**
- Daily homework problems
- Data entry corrections
- Resetting system passwords

Non-Routine Problems

- A **non-routine problem** is a problem that **is new or unfamiliar and requires logical reasoning and analytical thinking**.
- **Characteristics**
- No ready-made solution
- Enhances thinking ability
- Requires trial and error
- **Examples**
- Logical puzzles
- Debugging new software errors
- Research-based problems

Final Thought

The quality of your solutions depends on the quality of your thinking.

“A problem is not a barrier to success; it is an invitation to think, analyze, and grow.”